# Data Strcture

Padmaraj Nidagundi

# Four Types of Queues

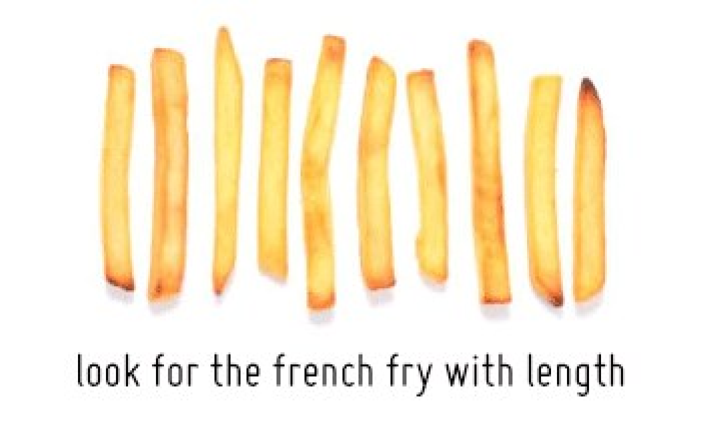
1. Linear Searching
2. Binary Searching
3. Sequential Searching
4. Interpolation Searching

# Linear Searching

Linear search is the simplest search algorithm in data structures. It iteratively checks each element in the collection one by one until a match is found. It's ideal for unsorted data structures, but it’s time complexity can be high for large datasets.

# Algorithm

Step 1 − Linear search will accept an array and a target value.
Step 2 − Start searching from the beginning of the array.
Step 3 − Check if that value equals the target:
If so, stop and return that values index.
If not, move onto the next element.
Step 4 − Repeat step 3 until all elements are checked. If target not found, return -1.


look for the french fry with length

# Binary Searching

Binary search is a highly efficient search method that works best with sorted data structures. It employs the "divide and conquer" approach, repeatedly dividing the search space in half until the desired element is located. This results in significantly faster searching for large datasets.

**Divide and Conquer** is a problem-solving strategy that involves breaking down a complex problem into smaller, more manageable parts, solving each part individually, and then combining the solutions to solve the original problem. It is a widely used algorithmic technique in computer science and mathematics.

# Algorithm

Step 1 − Select the middle item in the array and compare it with the key value to be searched. If it is matched, return the position of the median.

Step 2 − If it does not match the key value, check if the key value is greater than or less than the median value.
Either

Step 3 − If the key is greater, perform the search in the right sub-array; but if the key is lower than the median value, perform the search in the left sub-array.

Step 4 − Repeat Steps 1, 2 and 3 iteratively, until the size of sub-array becomes 1.

Step 5 − If the key value does not exist in the array, then the algorithm returns an unsuccessful search.